

Species Interaction Networks: A Macro-level Overview

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Themes

— — —

1. Why Networks?
2. Networks 101
3. Constructing Food Webs
4. Measuring Networks
5. Information about the Part
6. Trophic Cascades
7. Dynamic Models
8. Where to next?

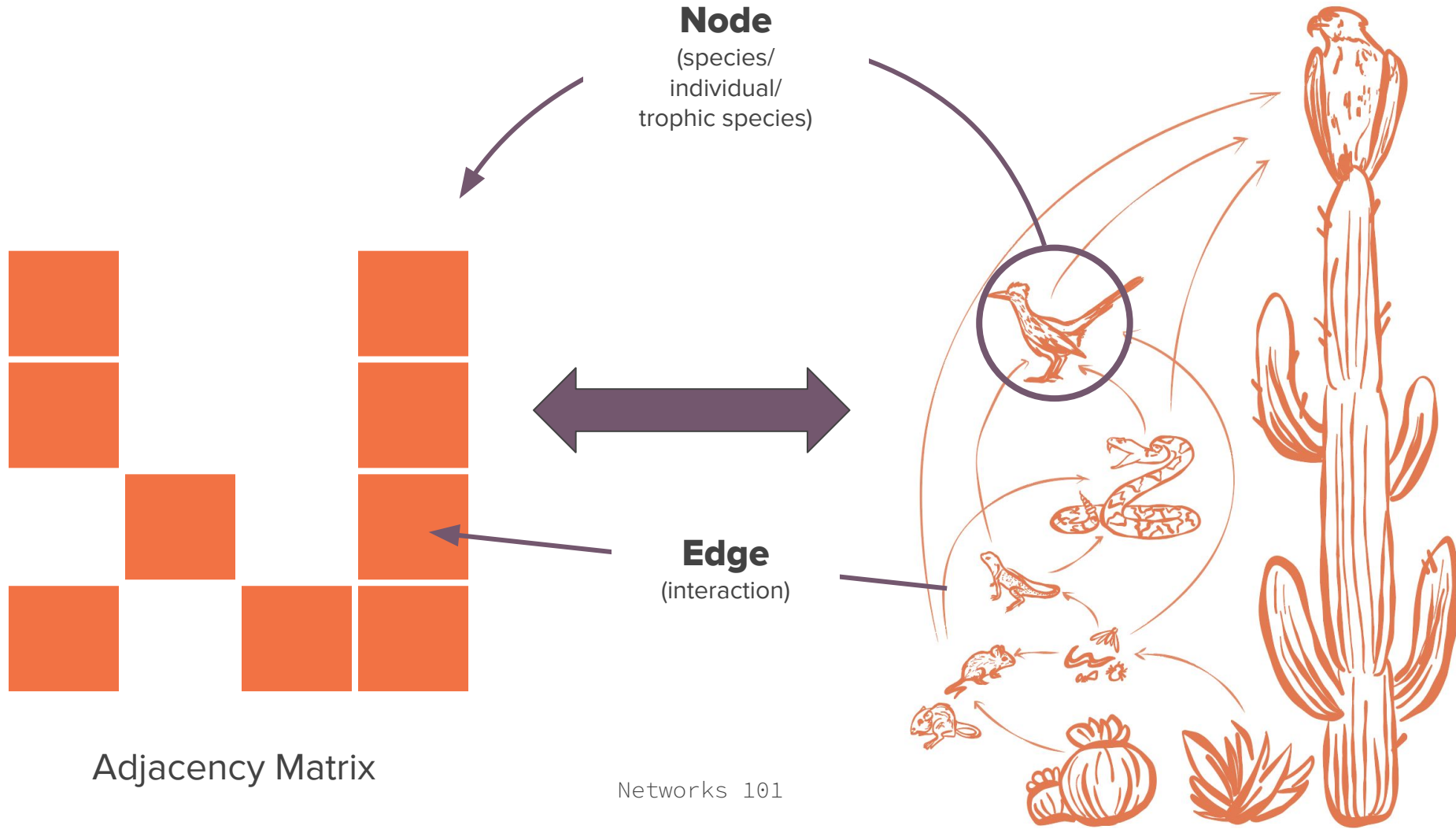
Why Networks?

What can we learn?

— — —

- Population -> Community -> Ecosystem
- How does a part fit into the whole
 - Invasive species (rewilding)
 - Extinctions
 - Diets
- Dynamics: no species is an island
 - Propagation of change/stress
 - Top-down/bottom-up control
 - Species persistence/co-existence (stability)

Networks 101



NODES

Species



Individuals



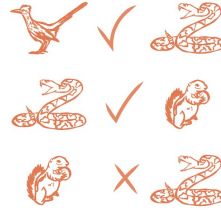
Trophic species

Same predators and prey



EDGES

Potential feeding links

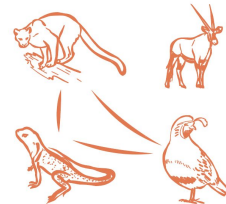


Feeding preference



Spatial associations

Co-occurrence networks



Quantifying an edge

- Binary
- Probabilistic
- Quantitative

Types of Networks

NON TROPHIC

Plant-pollinator, host-parasite, plant-herbivore

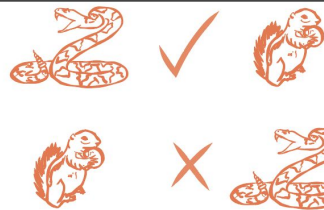


BIPARTITE

FOOD WEBS

METAWEB

UNIPARTITE



Who can eat who?

REALISED WEB



What does a predator choose to eat

Takeaways

It is possible to encode completely different processes within a network



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Constructing Food Webs

A two-body problem

1. A need for real world data
2. The development of reconstruction tools/approaches

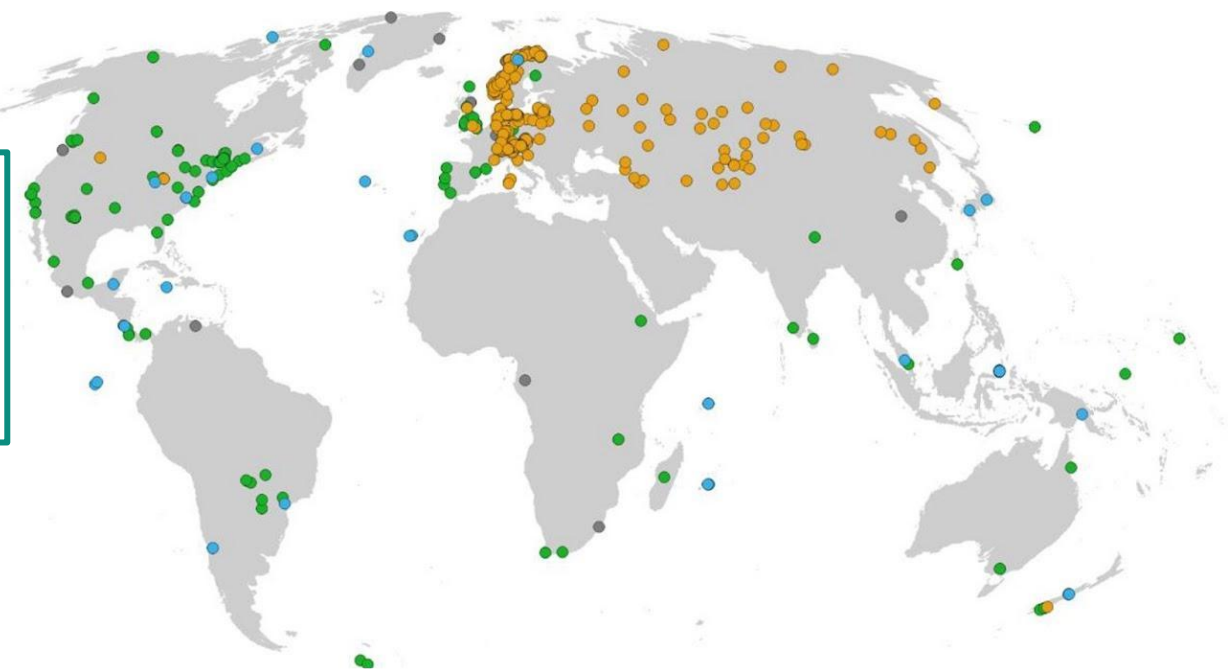


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The Challenge: Observing interactions is hard!

— — —

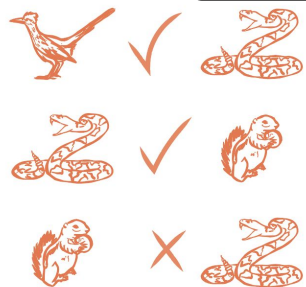
- Biases
 - Type
 - Location
- Methods
- Effort

- 
- mutualism
 - parasitism
 - predation
 - all networks

The Challenge (pt 2): 'Codification'

Strydom et al. (2026) EcoEvoRxiv

METAWEB



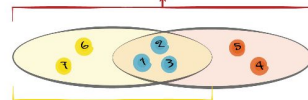
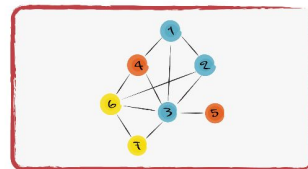
REALISED WEB



REPRESENTATION

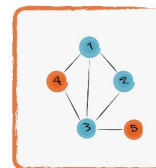
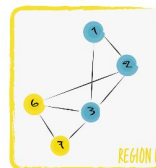
GLOBAL METAWEB

All feasible interactions for a collection of species



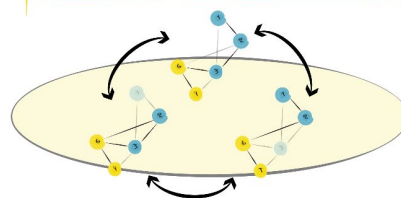
REGIONAL METAWEB

All feasible interactions for a co-occurring collection of species (i.e., interactions for a specific community)



REALISED NETWORKS

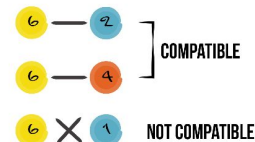
Interactions that occur within a community for a given time and place. These networks represent the dynamics between species in the communities



PROCESS



EVOLUTIONARY COMPATIBILITY



EVOLUTIONARY COMPATIBILITY



CO-OCCURRENCE



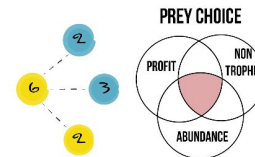
ABUNDANCE



PROFITABILITY

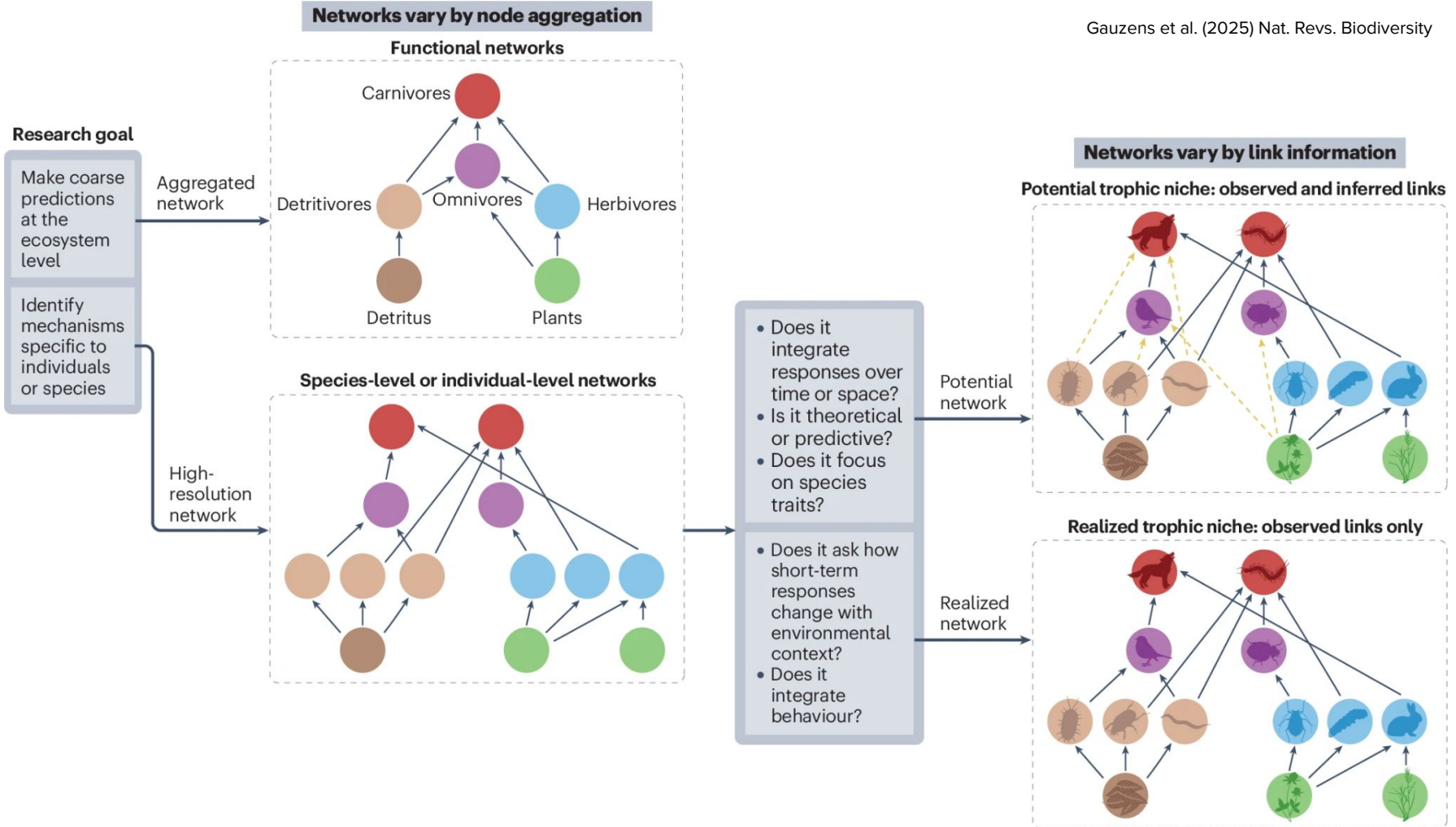


NON-TROPHIC INTERACTIONS



WHAT DETERMINES THE PRESENCE OF LINKS?

WHAT DETERMINES THE DISTRIBUTION OF LINKS?



Definitions impact tool development

A brief mud dinosaur interlude



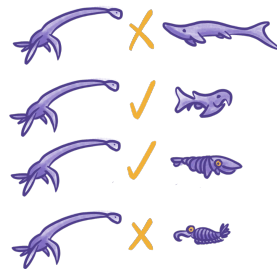
Hypothesis Embedding

— — —

Reconstruction encodes a set of assumptions using different types and sources of data

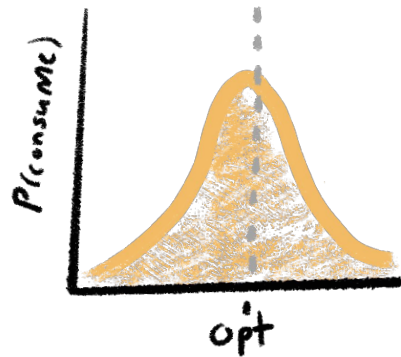
Mechanistic Models

Mechanistic processes.
Assess if pairwise interactions are feasible



Ecological Models

Embedding ecological theory and assuming network structure



Random Models

Assume that interactions between species are random and that the number of links is constrained. No *biological* structuring

Hypothesis Embedding

Mechanistic Models

Expert knowledge

Phylogenetic

Hierarchical

Binary classifiers

Graph embedding

Ecological Models

Cascade

Body mass scaling
(constrained)

Niche

Body mass scaling

Neutral

Diet models

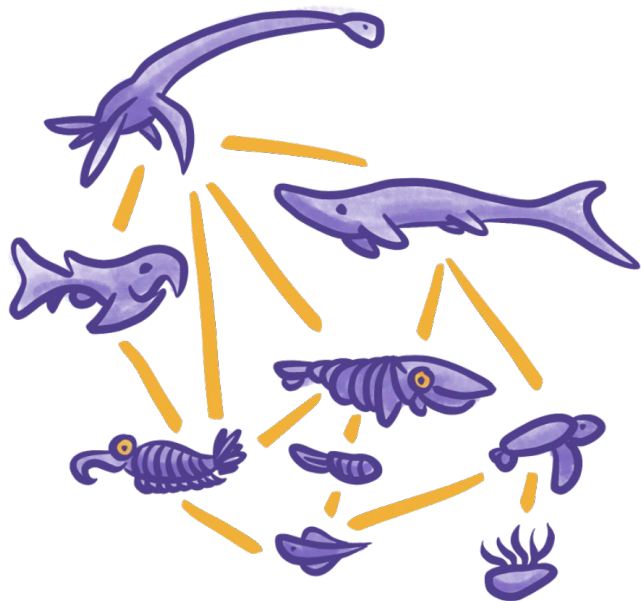
Statistical

Theoretical

Random Models

Erdős–Rényi

Null models



Hypothesis Embedding

Mechanistic Models

Expert knowledge

Phylogenetic

Hierarchical

Binary classifiers

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Ecological Models

Cascade

Body mass scaling (constrained)

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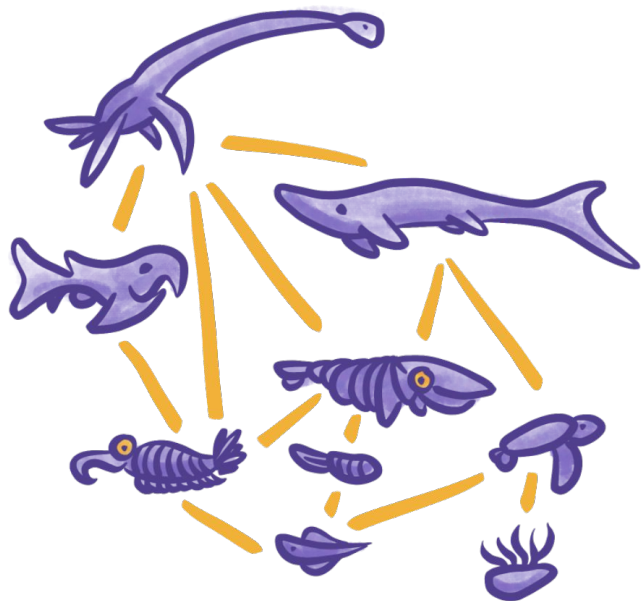
Statistical

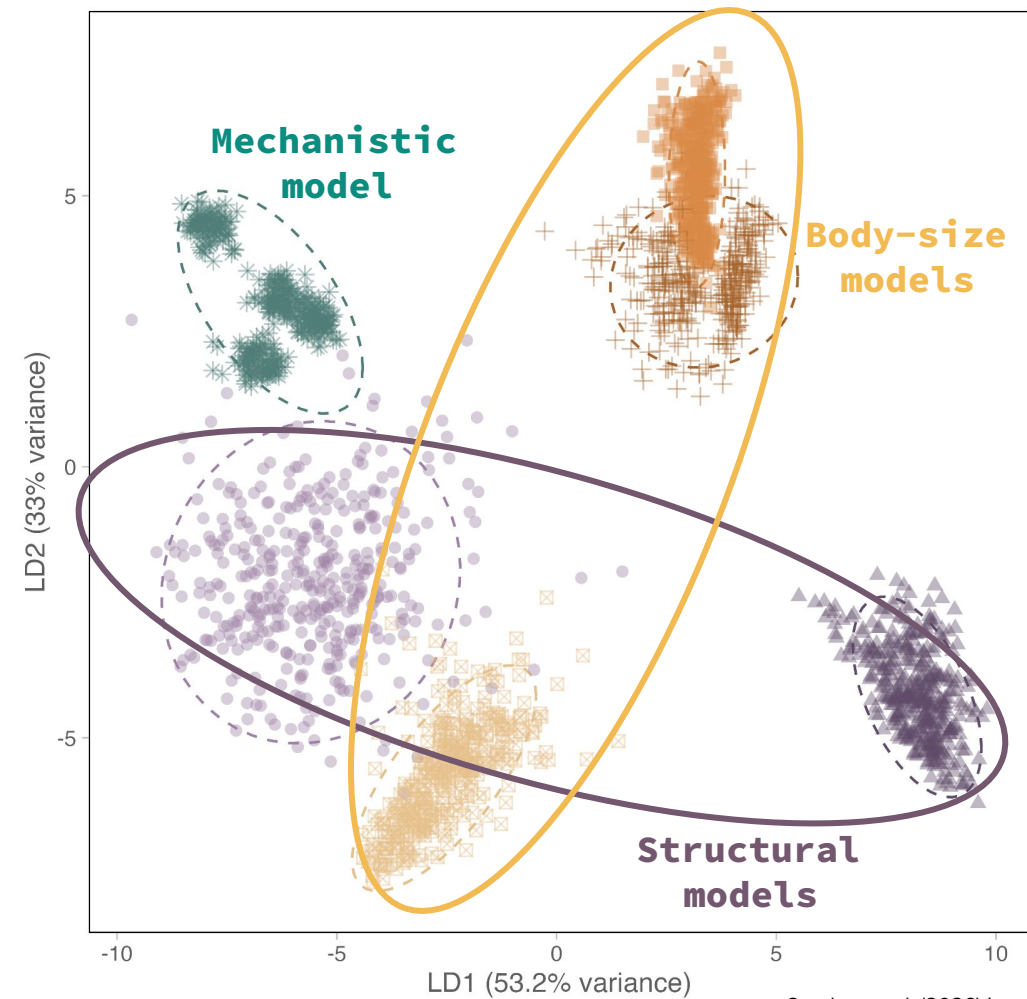
Theoretical

Random Models

Erdős–Rényi

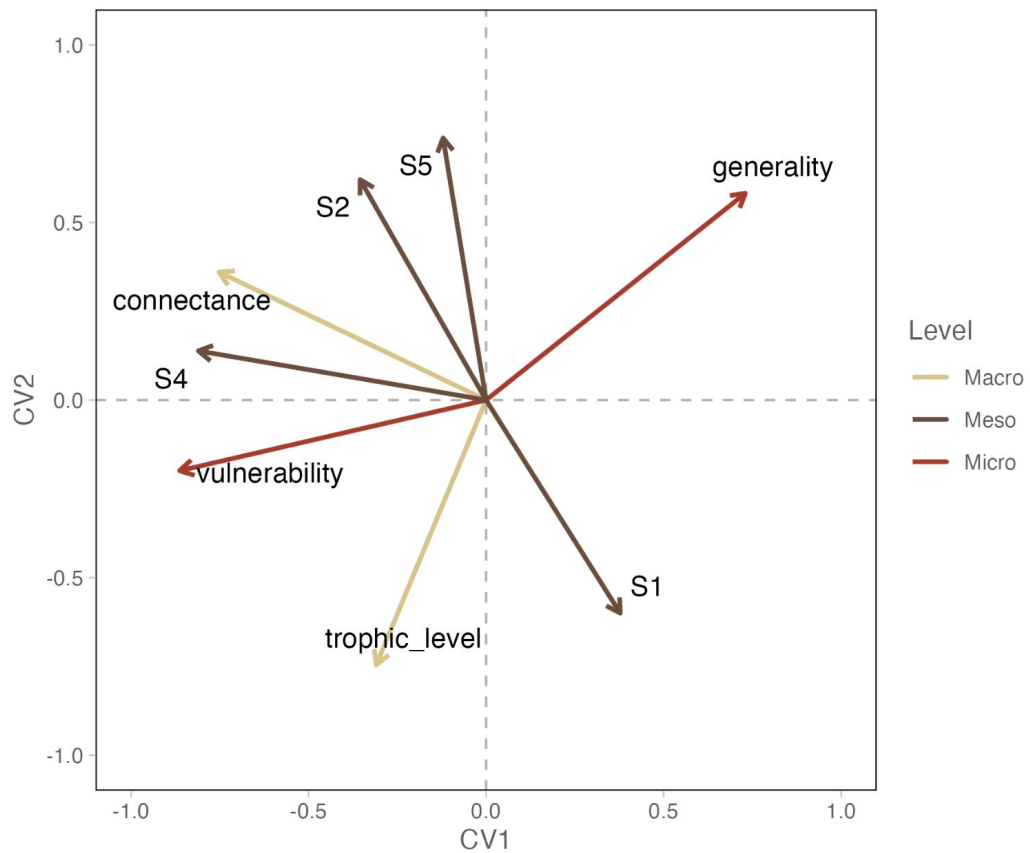
Null models





Strydom et al. (2026) In prep.



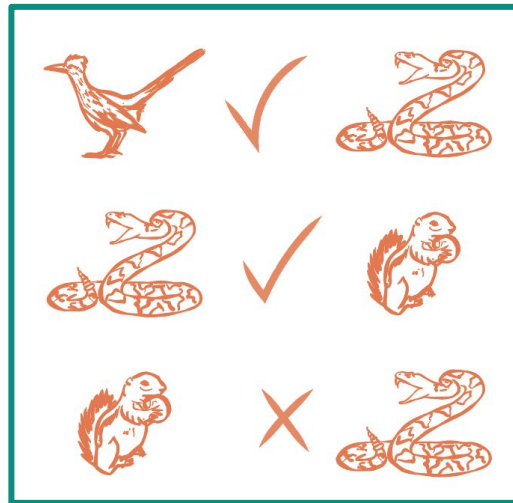


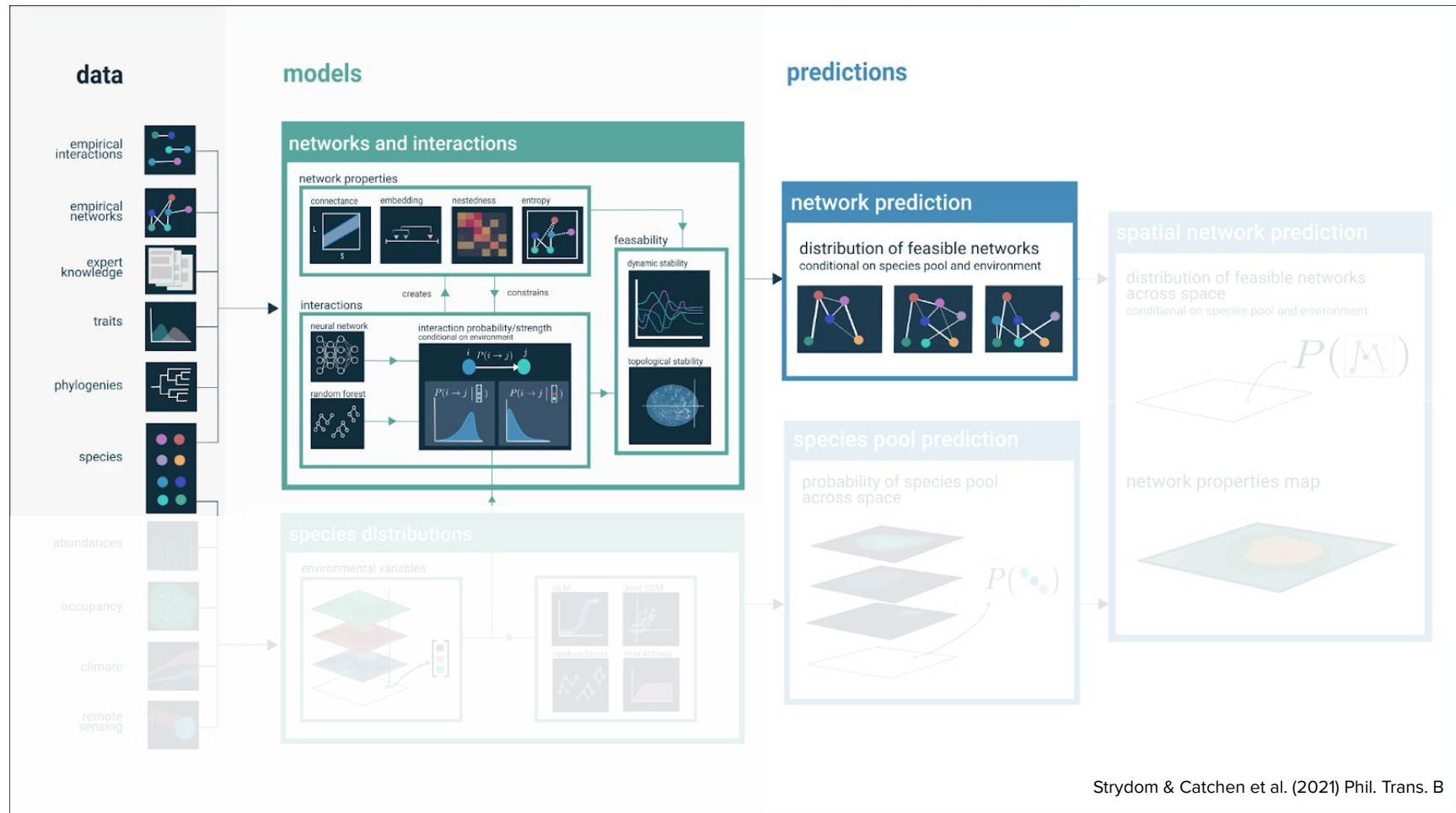
A more detailed overview of said reconstruction tools

Predicting Interactions: Mechanistic Models

— — —

- Construct **metawebs**
- Does sp_a have the capacity to eat sp_b ?
- Assumption that ‘feeding rules’ determine interactions
 - Rules can be defined *a priori*^[1]
 - Inferred from known networks^[2,3]
 - Use of ecological proxies (e.g., phylogeny)^[4,5]

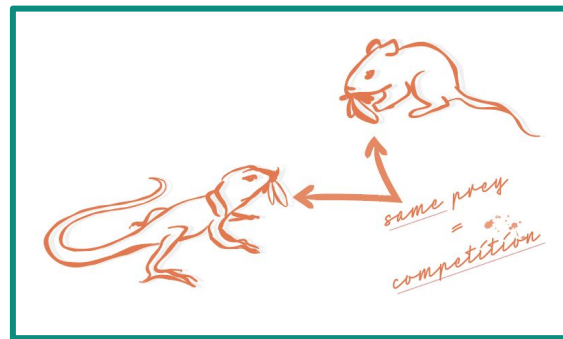




Strydom & Catchen et al. (2021) Phil. Trans. B

Predicting Interactions: Theory-driven Models

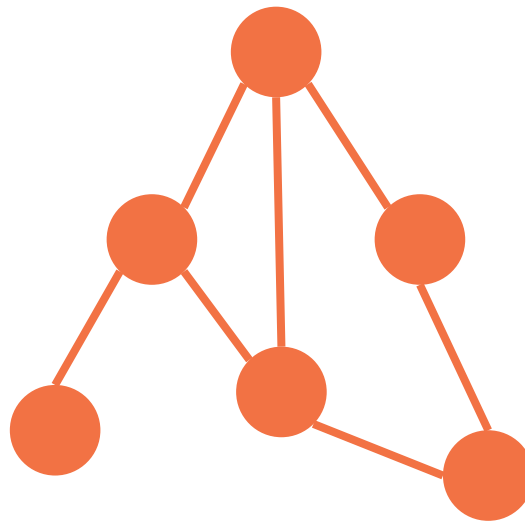
- Construct *realised webs*
- Reflect the realisation of feasible links
- Intersection of profitability, abundance, non-consumptive interactions
 - Foraging ecology^[6,7]
 - Allometry (body size)^[8]
- **Dynamic**



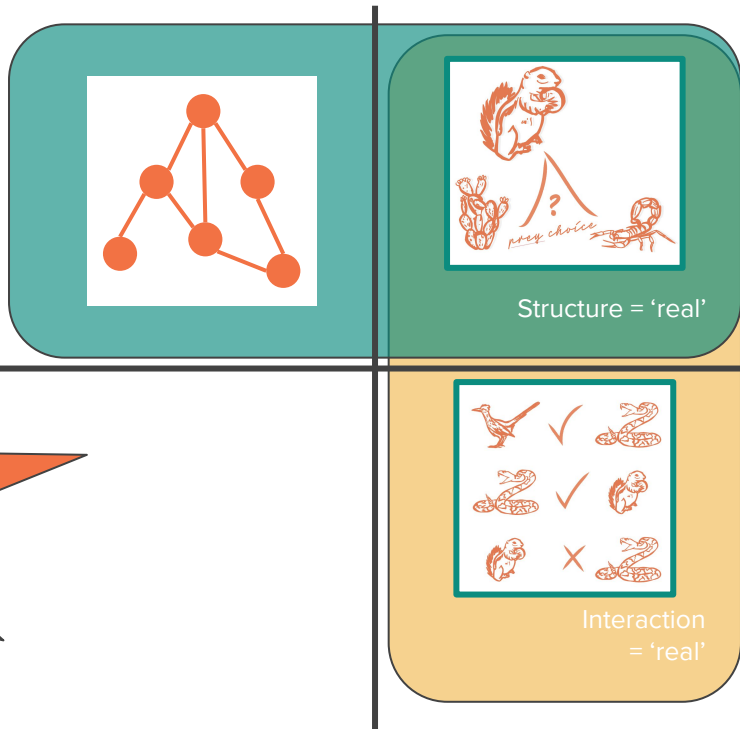
Predicting Structure

— — —

- Construct *topologies*
- Assumptions of expected structure of a network (distribution of links)
 - Random
 - Neutral (abundance)
 - Stochastic^[9]



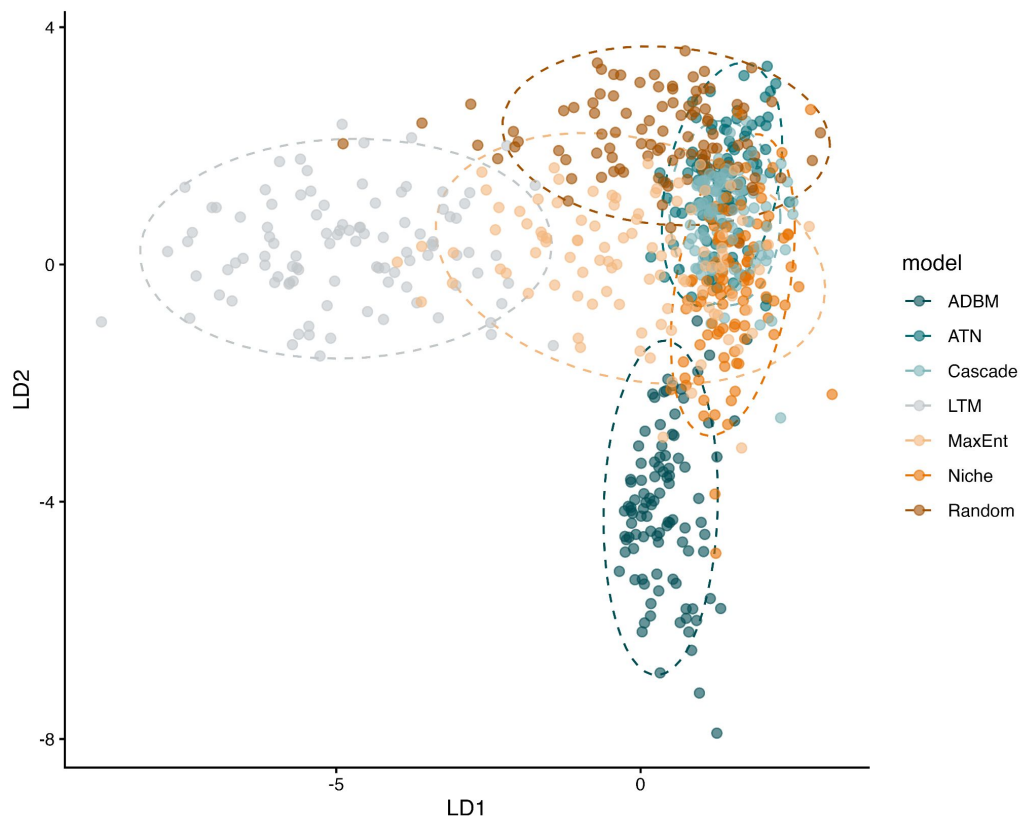
Predicting Networks: The Divergence



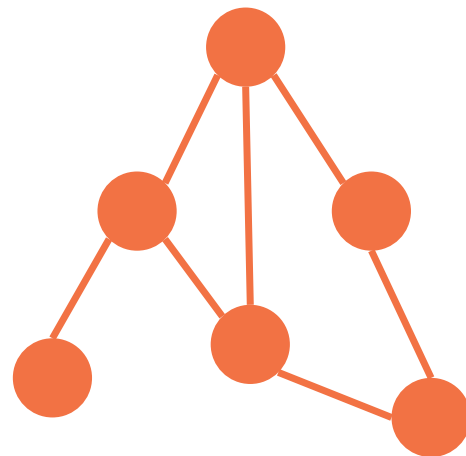
- Currencies
- Use: Theory v. applied
- Scale: Realised v. bigger picture
- 'Accuracy': Model v. reality

OK but who should we 'believe'?

Similar tools create similar networks



Well if you squint... All networks that you can synthetically create something with



What are the opportunities that different network representations afford us?

— — —

Metawebs	Realised webs	Topological webs

What are the opportunities that different network representations afford us?

— — —

Metawebs	Realised webs	Topological webs
• ‘Global overview’ • Who <i>could</i> eat who? • ‘Condition invariant’ • Full breadth - fundamental niche • On the ground applications? Data light, easy to obtain	• Here and now • Who <i>chooses</i> to eat who • Dynamic - depends on bigger community/environment context • ‘Realized niche’ • Structure realistic - actual energy flows/pathways && species info	• Useful in theoretical scenarios/contexts • Real world abstraction - operate in synthetic space • Structure = informative -> ecosystem level processes

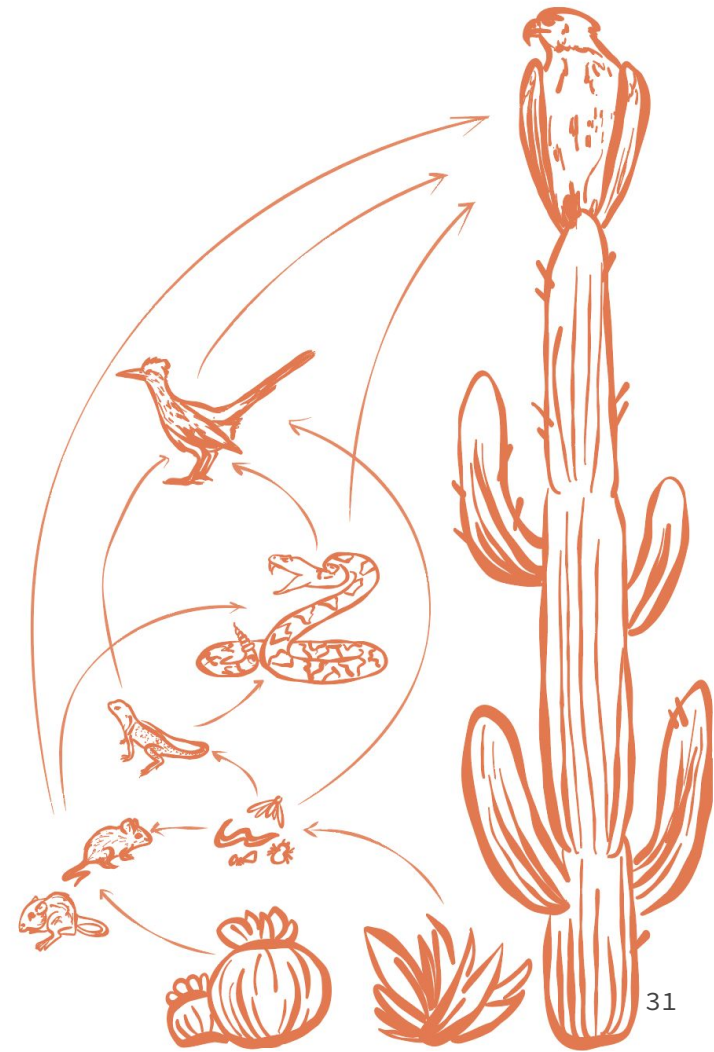
Takeaways

- Prediction vs Modelling
- Different families/approaches
- Different end uses



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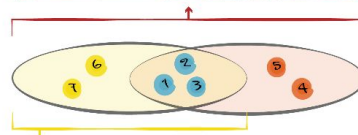
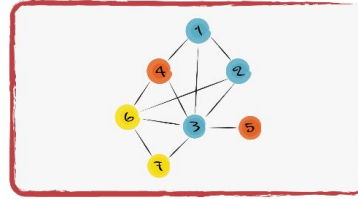
Species Interactions: Part 2



REPRESENTATION

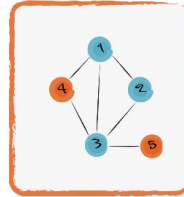
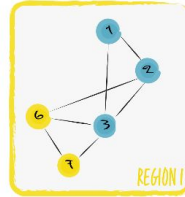
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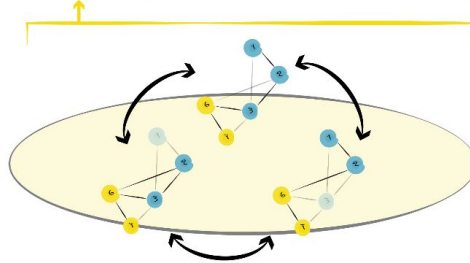
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REALISED NETWORKS

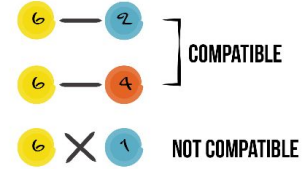
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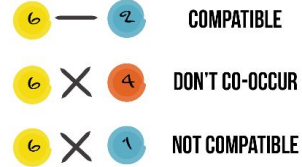
PROCESS



EVOLUTIONARY COMPATIBILITY



EVOLUTIONARY COMPATIBILITY



CO-OCCURRENCE



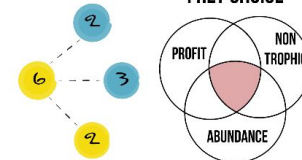
ABUNDANCE



PROFITABILITY



NON-TROPHIC INTERACTIONS



WHAT DETERMINES THE PRESENCE OF LINKS?

WHAT DETERMINES THE DISTRIBUTION OF LINKS?

Measuring Networks^[10]

It's not just about diets

1. What can we learn from network structure?
2. The role of species?



— — —

Macro

(Whole network)

- Connectance
- Diameter
- Trophic levels
- Nestedness
- Modularity
- Complexity
- Redundancy

Are certain properties only relevant for certain types of networks?

Micro

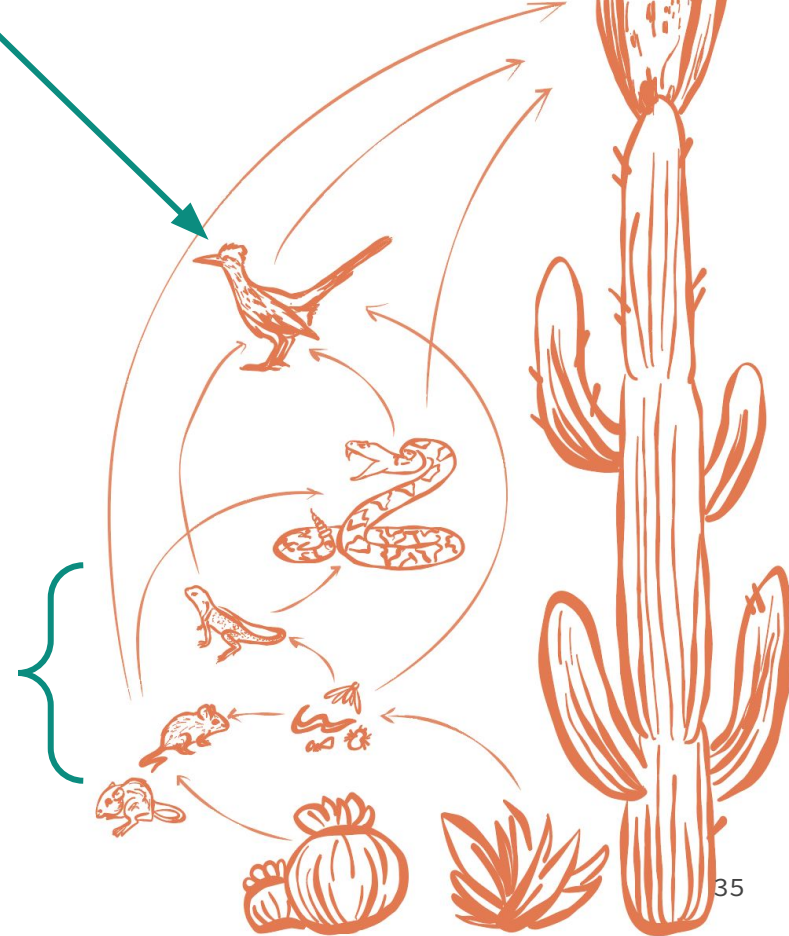
(Species)

- Specificity
- Centrality
- Degree
 - Generality
 - Vulnerability

Meso

(‘Units’)

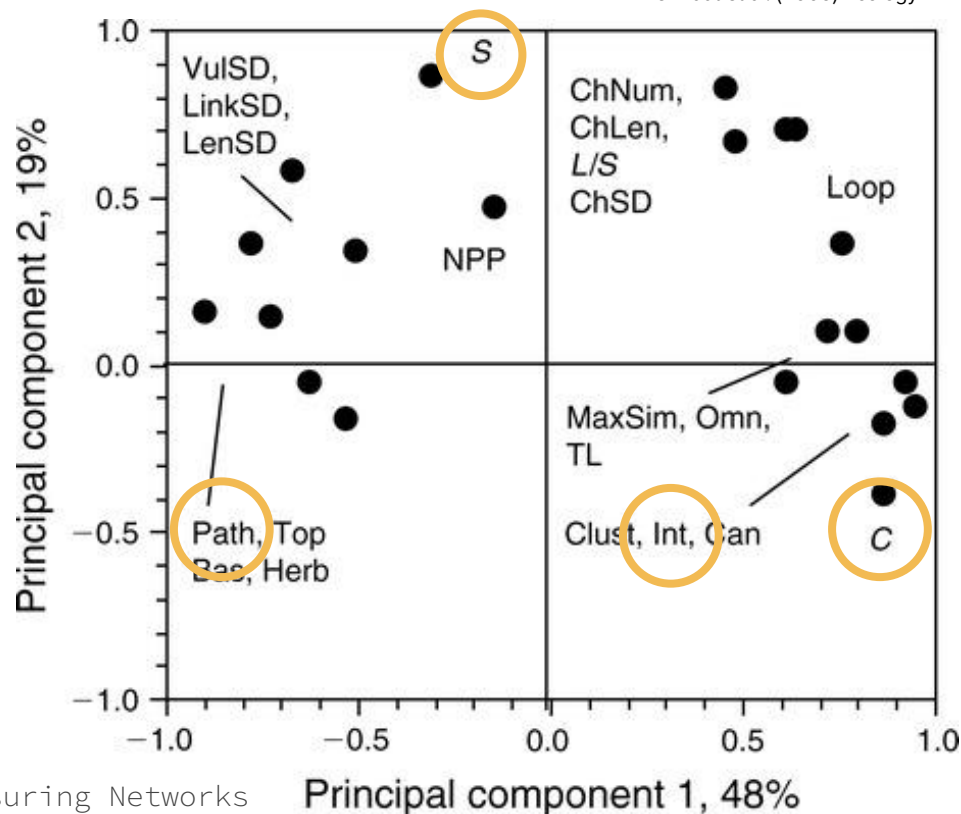
- Motifs^[Stouffer]
- Path length



Dimensions of 'network space'

Vermaat et al. (2009) Ecology

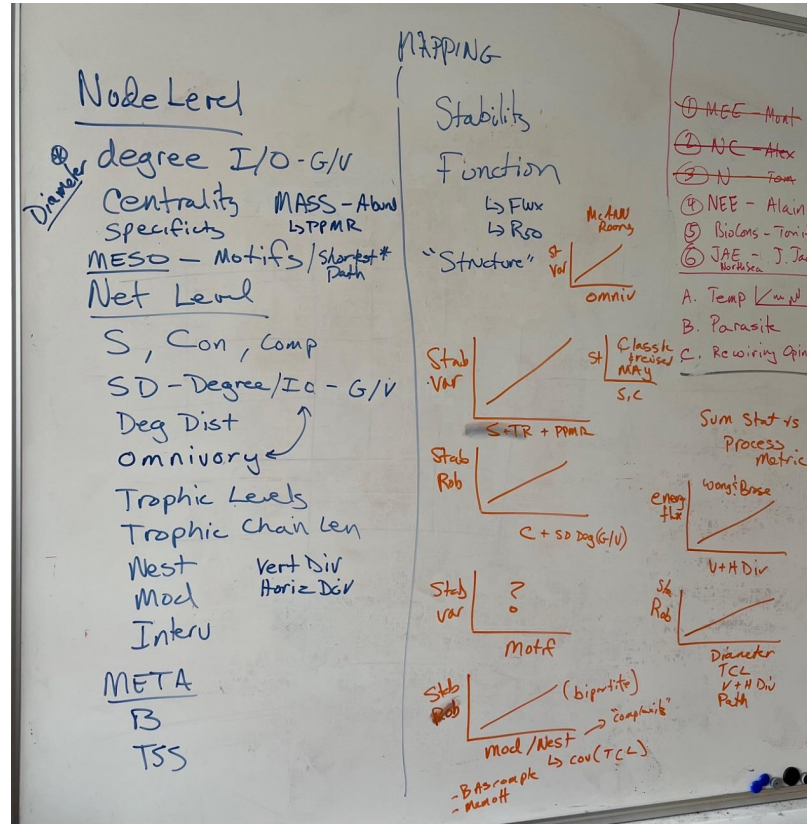
- What describes network space best?
- Is there a single metric we can use?



Linking form (structure) to function?

- Structure is the 'key' to understanding ecosystem function
- Remains a bit 'messy' - is there a single metric we can confidently ascribe to e.g. stability

Recommended reading: Thompson et. al., (2012) TREE



Takeaways

- Information at different scales
- Properties inform on behaviour of system

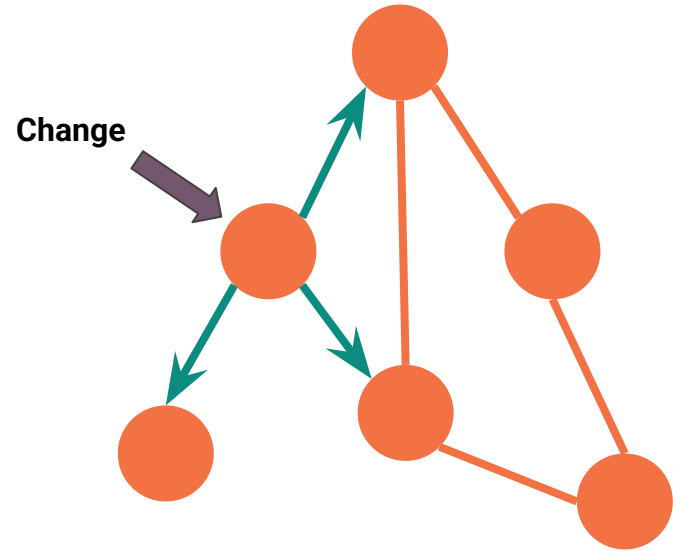


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Information about the part

Propagation of change

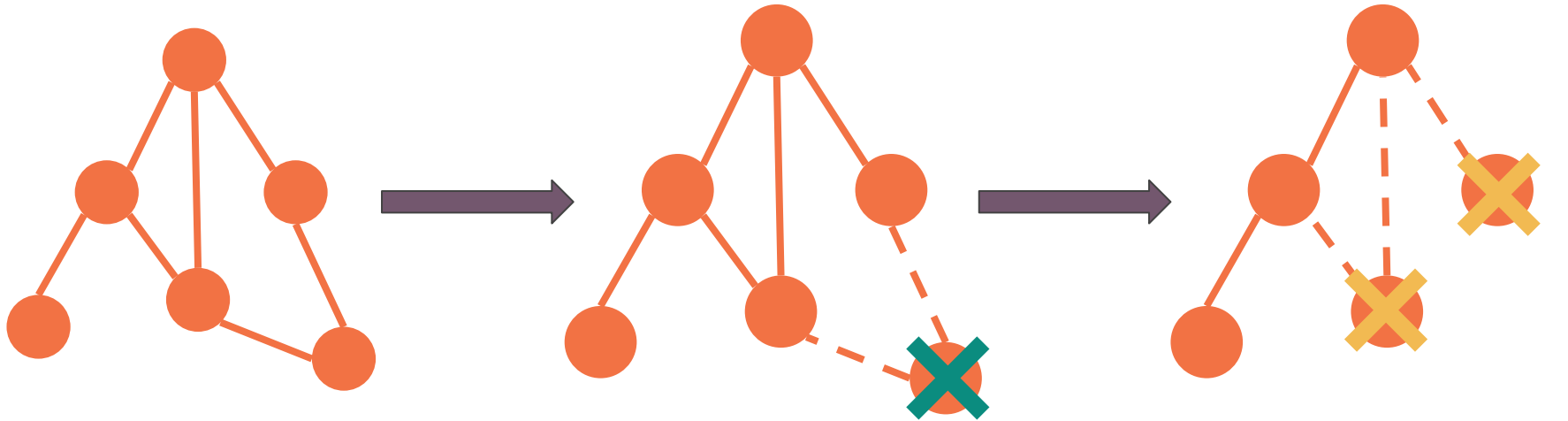
Changes in one node propagate through the network.



Extinctions

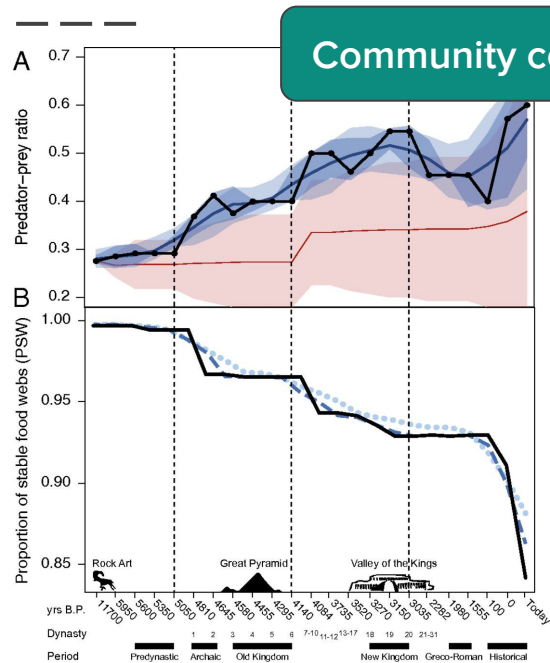
Primary Extinction

Secondary
Extinctions

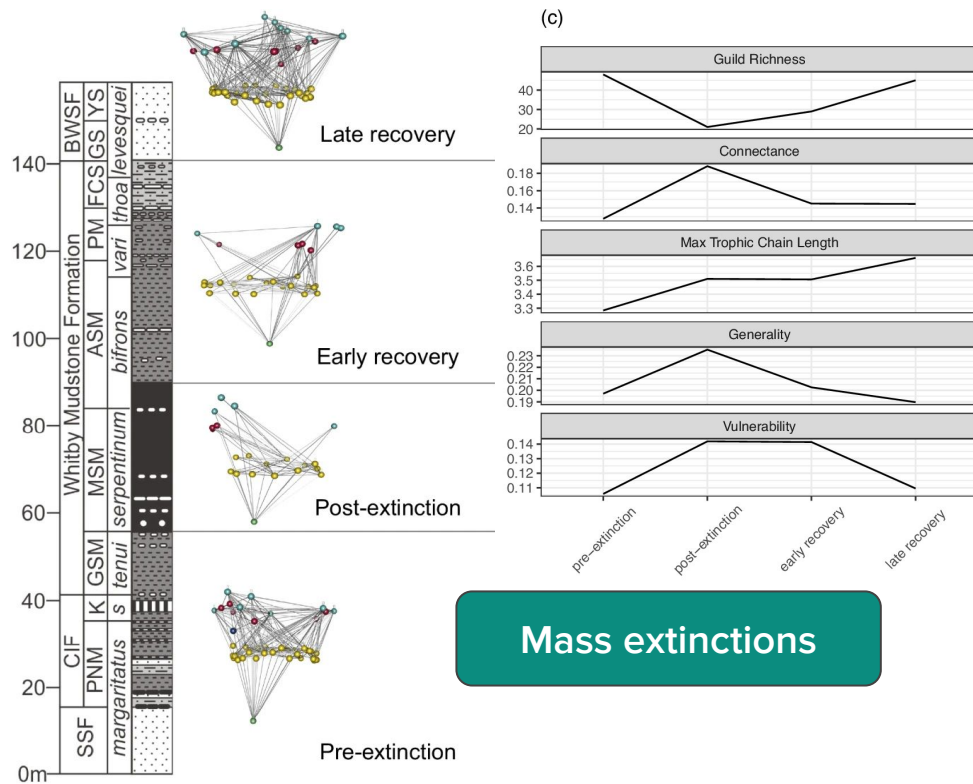


Information about the part

Extinctions



Yeakel et al. (2014) PNAS

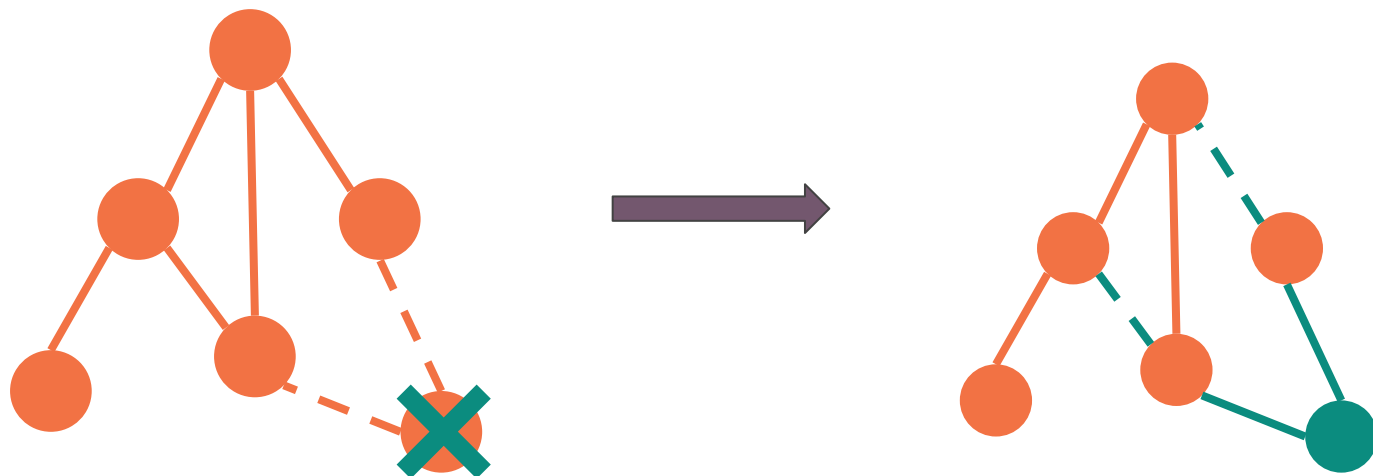


Dunhill et al. (2024) Nat. Comms.

Information about the part

'Invasions'

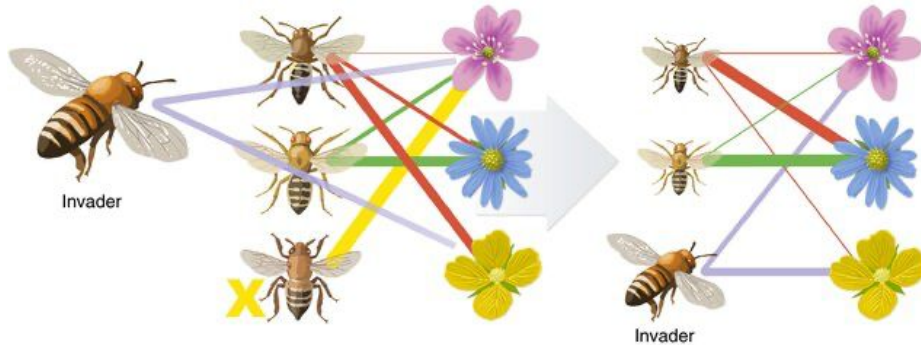
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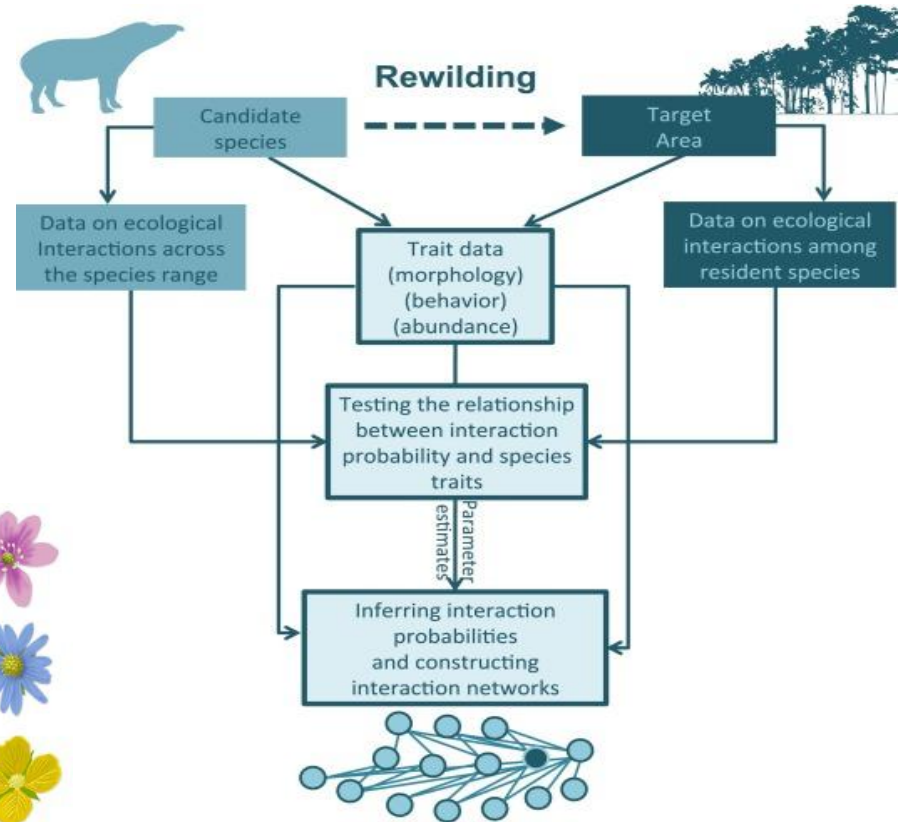
Information about the part

'Invasions'

- Adding species to a community
- What are the knock-on effects?



Valdovinos (2019) Eco. Lett.



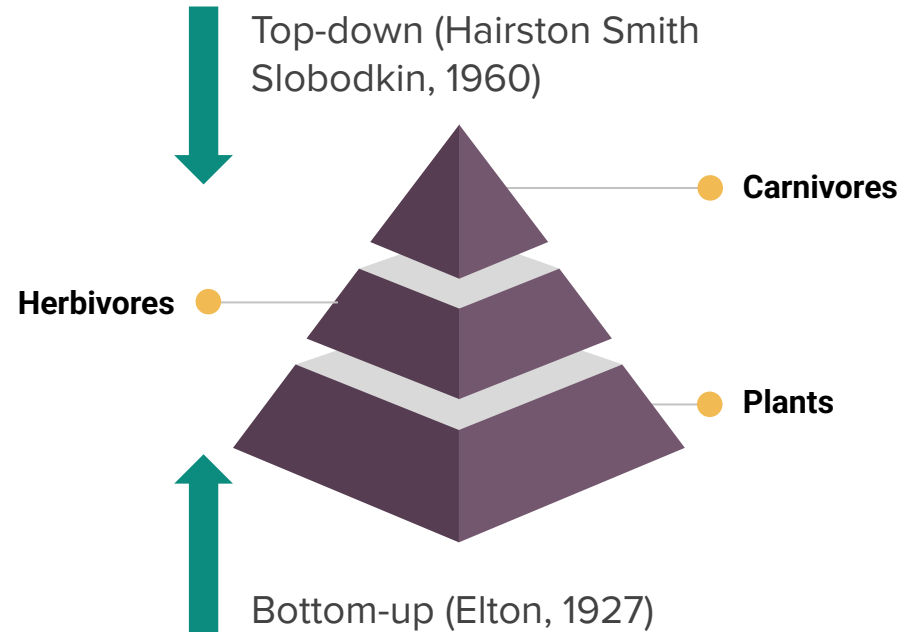
Pires. (2017) Perspectives Eco & Cons

Information about the part

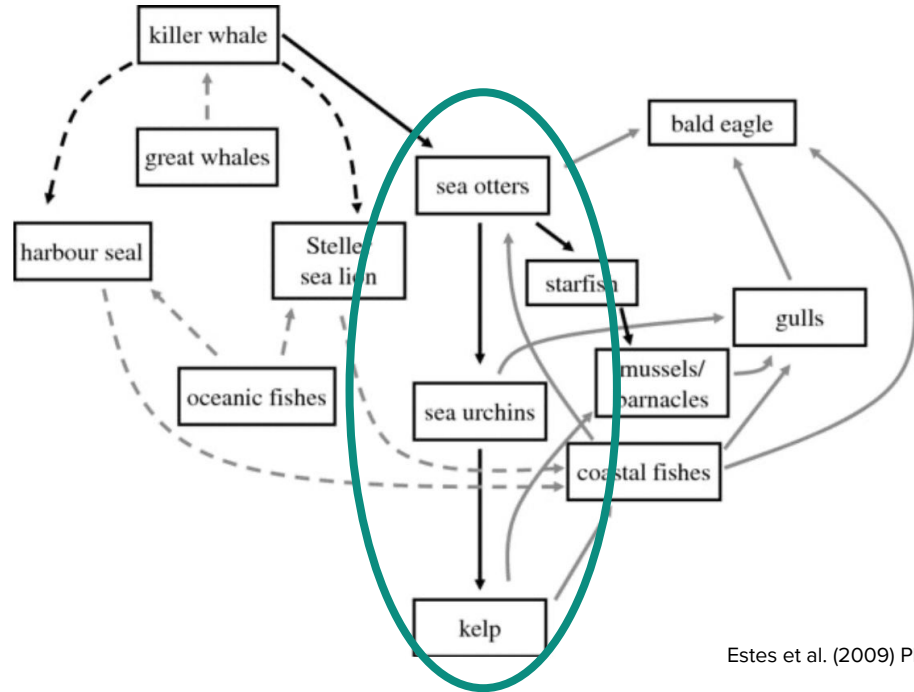
Trophic Cascades

Checks and balances

Who keeps who in line. Changes in the part leads to changes in the whole.



Kelp forests and sea otters



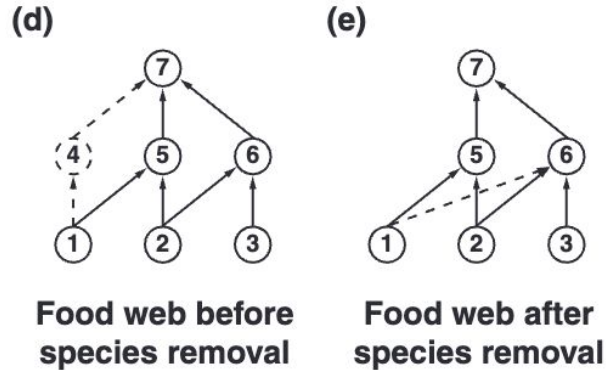
Estes et al. (2009) Phil. Trans. B

Foraging behaviour

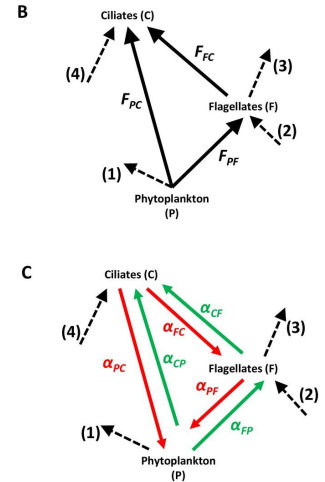
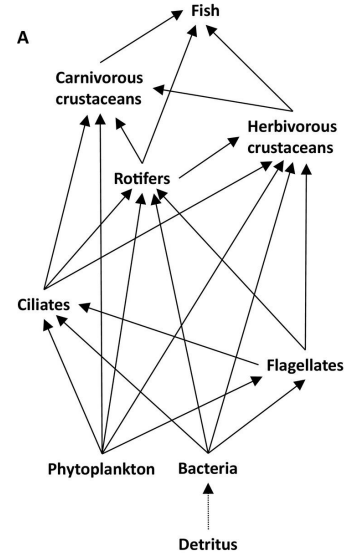


If dynamic why are networks stable?

- Rewiring
- Changes in interaction strength



Staniczenko et al. (2010) Eco. Lett.



Gaedke et al. (2025) Eco. Lett.

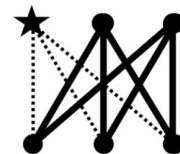
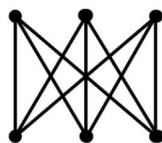
Rewiring

Resilience of the network to continue providing ecosystem function when losing ★ species

More resilient ↑
Less resilient ↓

BEFORE

AFTER

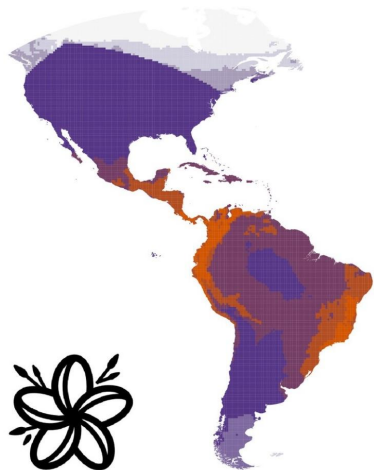


Rewiring pathway #1:
loss of a pairwise
interaction

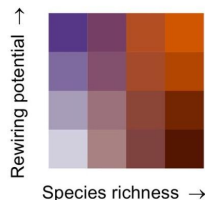
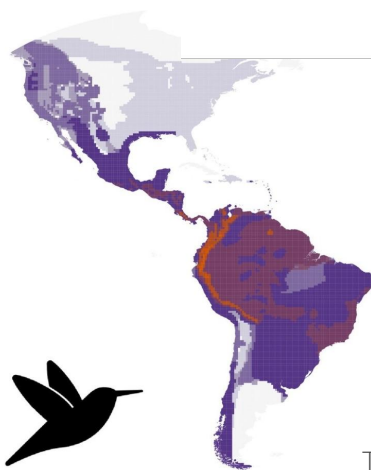
Rewiring pathway #2:
emergence of a new
pairwise interaction

Rewiring pathway #3:
alteration in the strength
of a pairwise interaction

a



b



Trophic Cascades

Takeaways

- Networks give multi-species insights on populations
- Populations are dynamic but networks remain stable



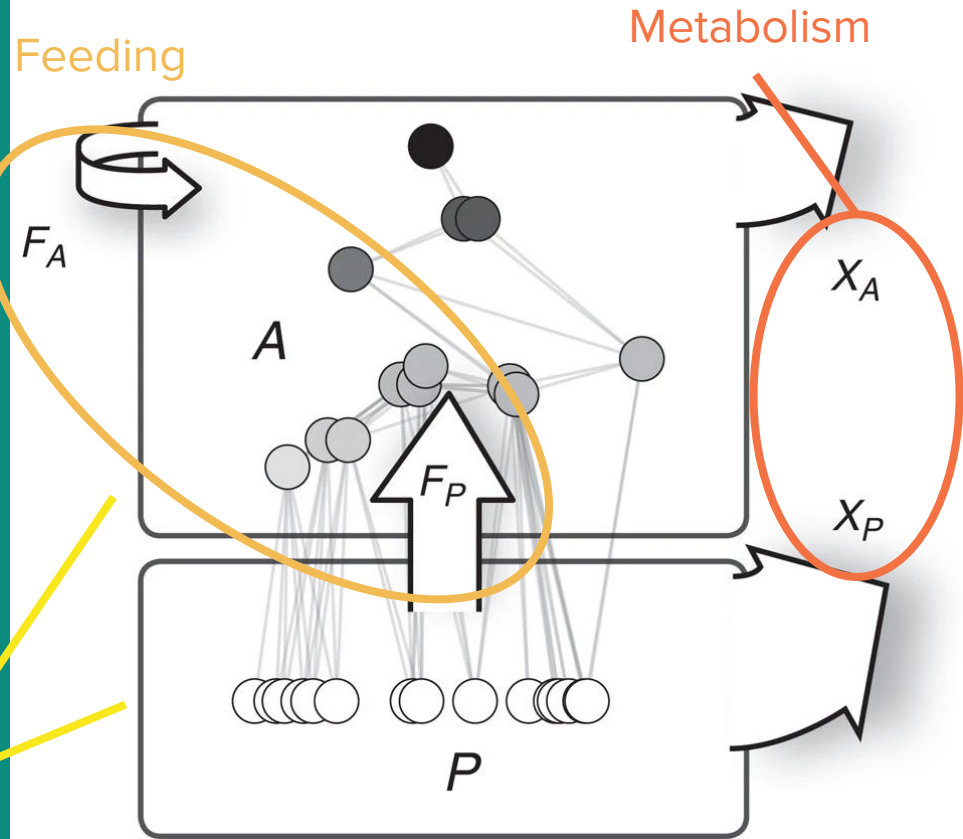
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Dynamic Models

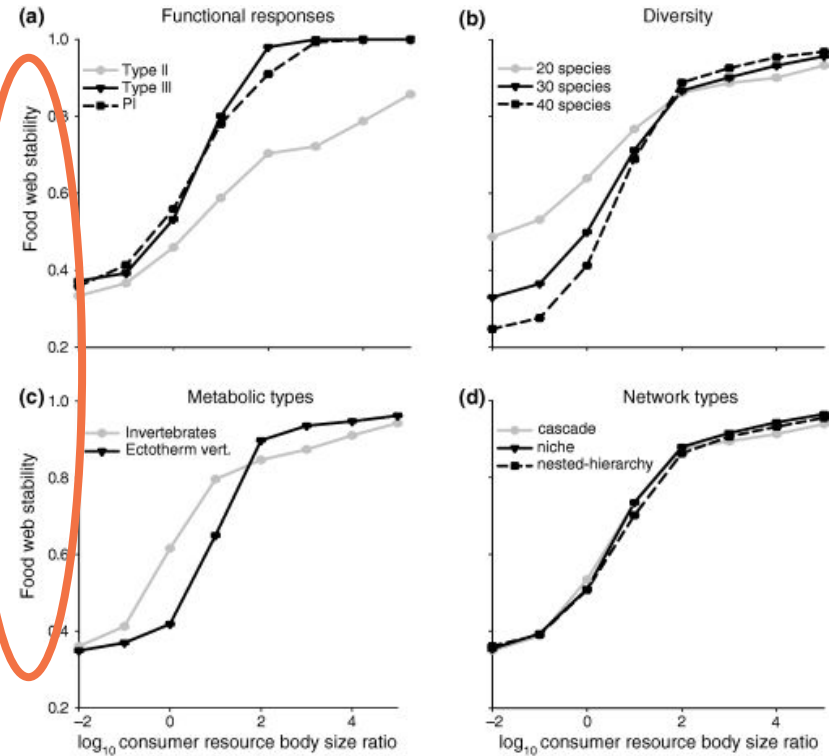
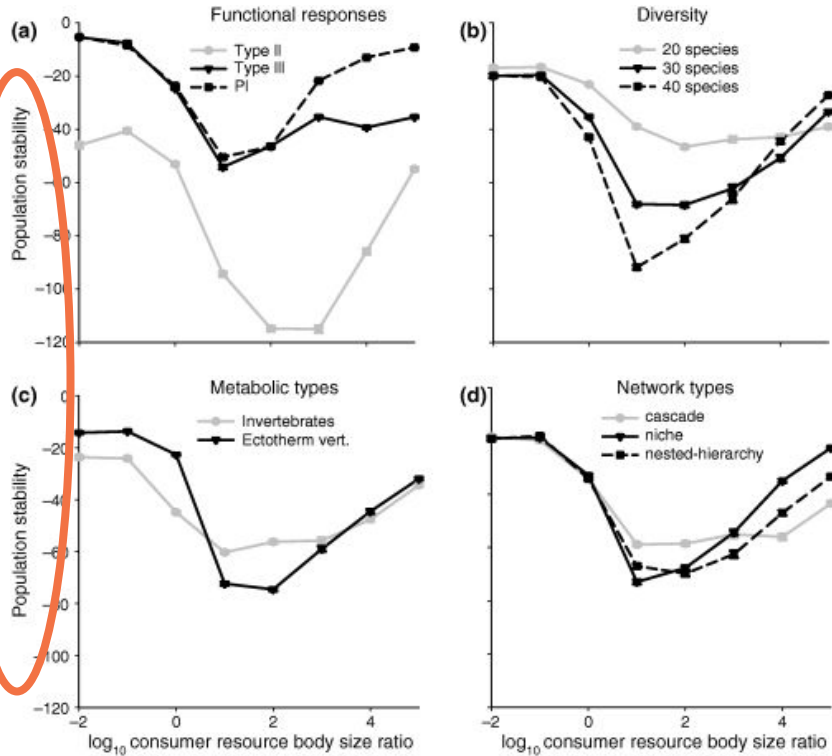
Lotka-Volterra plus

Bioenergetic Food Web Models

Biomass



A step towards understanding stability



Dynamic Models

**Where to
next?**

Methodological Development

— — —

- Estimating both feasibility and realisation of links
- Incorporating interaction strength
- Appropriate unit of scale?
- How do we validate our models or predictions?
- Making networks more tractable in an applied space - what can 'boots on the ground' conservation use?
- Networks in other facets of ecology -> joint SDMS, biogeography

Theoretical development

— — —

- Reinforcing old questions in ecology: coexistence, top-down v. bottom up
- Scaling up the stability-diversity-productivity discussion to complex communities
- Novel communities under climate change - the role of 'invasions' and re-wiring
- Combining socioeconomic and biological networks

References

1. Morales-Castilla, Ignacio, Miguel G. Matias, Dominique Gravel, and Miguel B. Araújo. 2015. "Inferring Biotic Interactions from Proxies." *Trends in Ecology & Evolution* 30 (6): 347–56. <https://doi.org/10.1016/j.tree.2015.03.014>
2. Stock, Michiel, Timothée Poisot, Willem Waegeman, and Bernard De Baets. 2017. "Linear Filtering Reveals False Negatives in Species Interaction Data." *Scientific Reports* 7 (April): 45908. <https://doi.org/10.1038/srep45908>.
3. Poisot, Timothée, Marie-Andrée Ouellet, Nardus Mollentze, Maxwell J. Farrell, Daniel J. Becker, Liam Brierley, Gregory F. Albery, Rory J. Gibb, Stephanie N. Seifert, and Colin J. Carlson. 2023. "Network Embedding Unveils the Hidden Interactions in the Mammalian Virome." *Patterns* 4 (6): 100738. <https://doi.org/10.1016/j.patter.2023.100738>.
4. Caron, Dominique, Luigi Maiorano, Wilfried Thuiller, and Laura J. Pollock. 2022. "Addressing the Eltonian Shortfall with Trait-Based Interaction Models." *Ecology Letters* 25 (4): 889–99. <https://doi.org/10.1111/ele.13966>.
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